

Universal Indicatrix in my words CHARACTERIZATION

Here is the verbatim transcription of the text contained in the provided images:

Digital Text Messages

Source 1 & 2: "Ok. One step. Drop this into an LLM, ask if to try. Either it is **plausible** or not

1. Tell LLM **neutral assessment** first, **derivation** second, **empirical matching** third.
2. tell it this **coordinate system** is enough to **derive physics** just by pinning **scaling to time** and constraining **rotations to small harmonics**, and using **solids of rotation** to approximate the shape of any field, constraining to **experimental measurements**. Reverse to derive the function of a **physical law** from measurements. I already tried but didn't have the coordinate system. Worked out. I think as a coordinate system, it's approximated by the **optical mineralogy** concept of the '**optical indicatrix**'. **Moire patterns** are similar. **System: four orthogonal spacelike axes with 4d unit hypersphere**. Superimpose that same setup on itself and allow the second to rotate about the origin. Now **fixed points trace any 3d trajectory**, add **scaling for time**, sweeping for **3d surfaces**
3. Ask LLM to try to derive an **analogue** of any well defined **physical law** or system, in these terms, or use my discord for attempted approximations to use as a starting point."

Handwritten Notebook Pages

Source 3:

- (Diagram of a sphere with intersecting axes and arrows)
- "Set up a **unit** ~~[crossed out text]~~ **hypersphere** ~~[crossed out text]~~ on **4 orthogonal spacelike axes**."
- "Set up a **second set** at overlapping axes defining a **2nd unit hypersphere**."

Source 4:

- "Allow the axes defining the unit sphere to **rotate freely** about the origin."
- "Add a **scaling factor** to the rotational axes to create a **mapping function** with arbitrary flexibility such that ~~[crossed out: define]~~ an **intersection** of ~~[crossed out: the]~~ a **ray** ~~[crossed out: traced]~~ through the origin defines ~~[crossed out text]~~ a **coincident point on both hyperspheres**."
- "~~[crossed out text]~~ But the point ~~[crossed out text]~~ is **fixed** to each sphere, so

if you rotate one the **vectors diverge**, the second now can **trace out any path** on the spheres, ie **hold one point still**"

Source 5:

- "use the **rotation** of the sphere ~~[crossed out: as]~~ as a ~~[crossed out text]~~ **control function** for ~~[crossed out text]~~ a **trajectory** of the ~~[crossed out: moving]~~ point **wrt** [with respect to] the ~~[crossed out text]~~ ~~[crossed out: fixed]~~ ~~[crossed out: point]~~ ~~[crossed out: the]~~ '**inner**' **fixed point**. ~~[crossed out: This defines a]~~"
- "**Integrate over all rotations** to write the full ~~[crossed out text]~~ **position system in 3D**... this function gives ~~[crossed out text]~~ **any possible path** in ~~[crossed out text]~~ **space**. **Scaling** gives all possible **positions in time**."
- "This allows **full 4-dimens'l mapping** of any point's path ~~[crossed out text]~~ **Solids of rotation** of the ray give all possible **geometric shapes**"

Source 6:

- "~~[crossed out text]~~"
- "• **Scaling** set to **linear expansion**"
- "• **Rot** [Rotation] limited to **small harmonics**"

Source 7:

- "Now ① we have the **coordinate system**."
- "② we have the **trajectory function**. (**3 rotation axes, 1 scaling axis**)"
- "③ **solids of rotation** about the ray"
- "~~[crossed out scribbles]~~"
- "**= fully parameterized grid + steering** for all **3D geometry** and **timelike transformations**."
- "Treat ~~[crossed out text]~~ **scaling function** as ~~[crossed out text]~~ **linear time expansion**."
- "Treat **solids of rotation** as models for **4d objects**. Constrain rot. to **small** [cut off: harmonics]."